

KELT-23Ab

R-2079

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_260322 · created 2026-08-09T23:57:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461119.6203 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.103 +/- 0.032
Transit depth (Rp/Rs)^2	1.06 +/- 0.67 [%]
Orbital Inclination (inc)	84.44 +/- 1.82
Airmass coefficient 1 (a1)	0.9093 +/- 0.0074
Airmass coefficient 2 (a2)	0.065 +/- 0.021
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	#3 - [494, 458]
Optimal Method	PSF photometry
Transit Duration (day)	0.026 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.103 ± 0.032 vs 0.132 (deviation 0.71σ)
Duration fit vs published	0.026 d vs 0.0954 d (deviation 4.53σ)
Mid-time O–C	-3249.9 min
Depth SNR	2.5
Residual scatter	0.81 %

Light curve

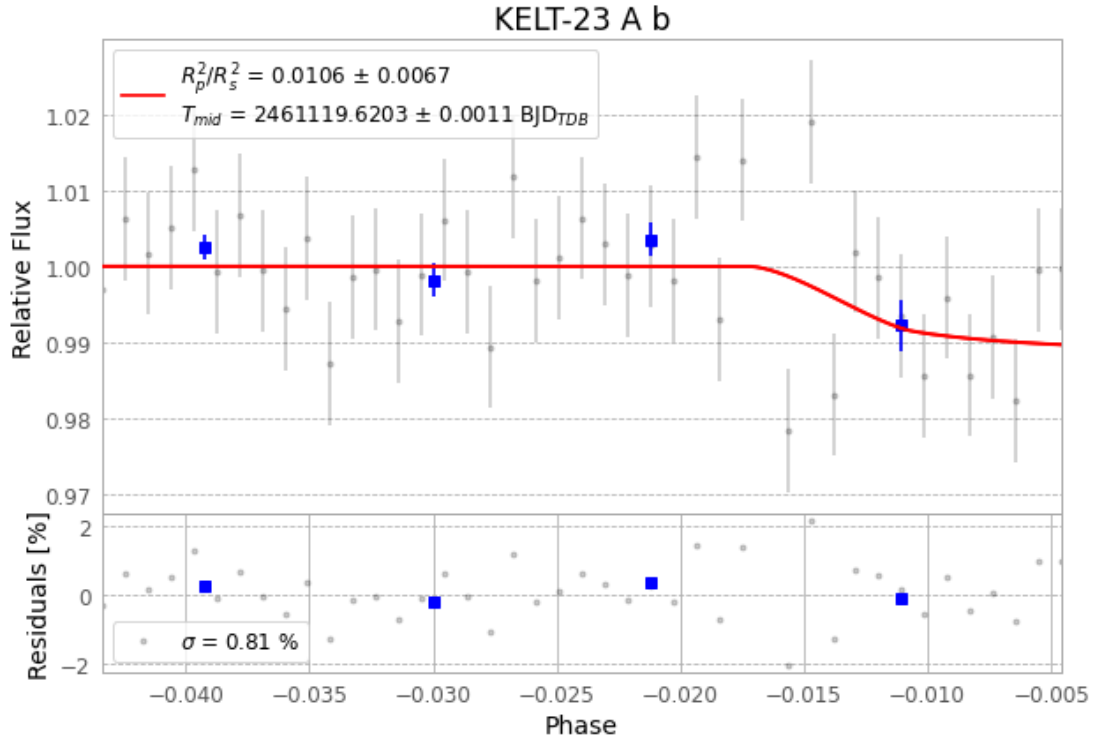


Figure 1 — FinalLightCurve_KELT-23 A b_2026-03-21.png (SHA-256 d55ad313180e6789...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ea1327fb6d12499b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

43 FITS frames (28 MB), KELT-23A260322063615.FITS → KELT-23A260322084216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-260322.log (e05b63629ab4...), FinalLightCurve_KELT-23 A b_2026-03-21.png (d55ad313180e...), FinalLightCurve_KELT-23 A b_2026-03-21.csv (235c3e6e56a7...)

Compute resources

Wall time 157.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:57Z.